

Deepfake Video Detection

Frame-Level CNN Classifier

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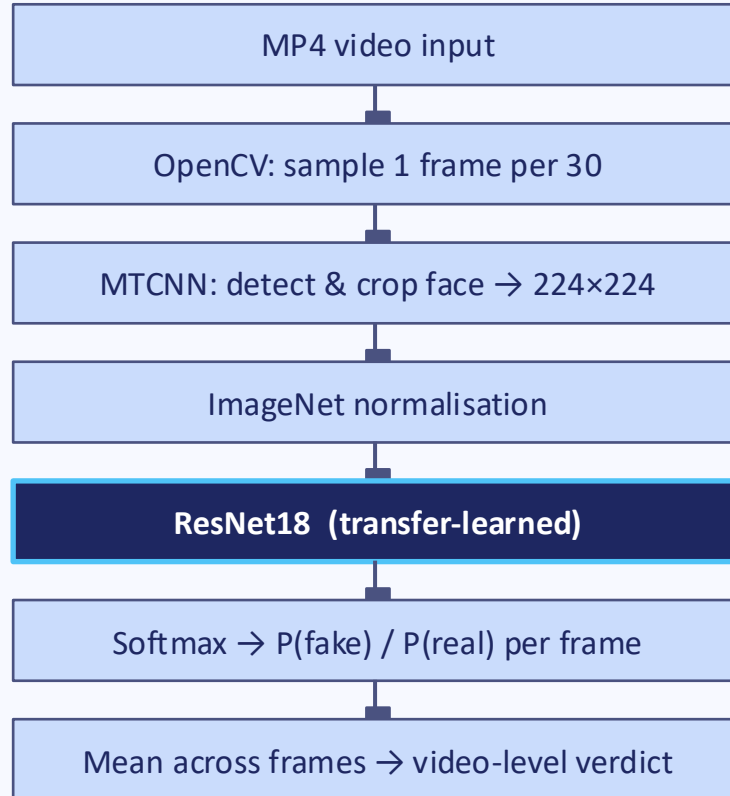
Problem & Motivation

- **What are deepfakes?**
 - Synthetically manipulated videos in which a subject's face is replaced or altered using generative AI, typically undetectable by an untrained observer.
- **Why detection matters**
 - Deepfake content poses measurable risks to public trust, journalistic integrity, and personal identity — several high-profile cases have already driven regulatory attention.
- **Project aim**
 - Build a frame-level binary classifier that scores each sampled face crop as REAL or FAKE, aggregating scores to produce a video-level verdict.

Background & Related Work

- **Key datasets**
 - FaceForensics++ (Rössler et al., 2019) and the DeepFake Detection Challenge (Dolhansky et al., 2020) are the standard benchmarks for this task.
- **Detection strategies**
 - Frame-level classifiers (this project) treat each face crop independently. Temporal models — 3D CNNs or LSTMs over frame embeddings — exploit inter-frame inconsistency but require greater computational resource.
- **Common backbones in the literature**
 - XceptionNet (Rössler et al.), EfficientNet, and ResNet variants are frequently compared. ResNet18 provides a compact, well-understood baseline suitable for a time-bounded undergraduate project.

System Architecture



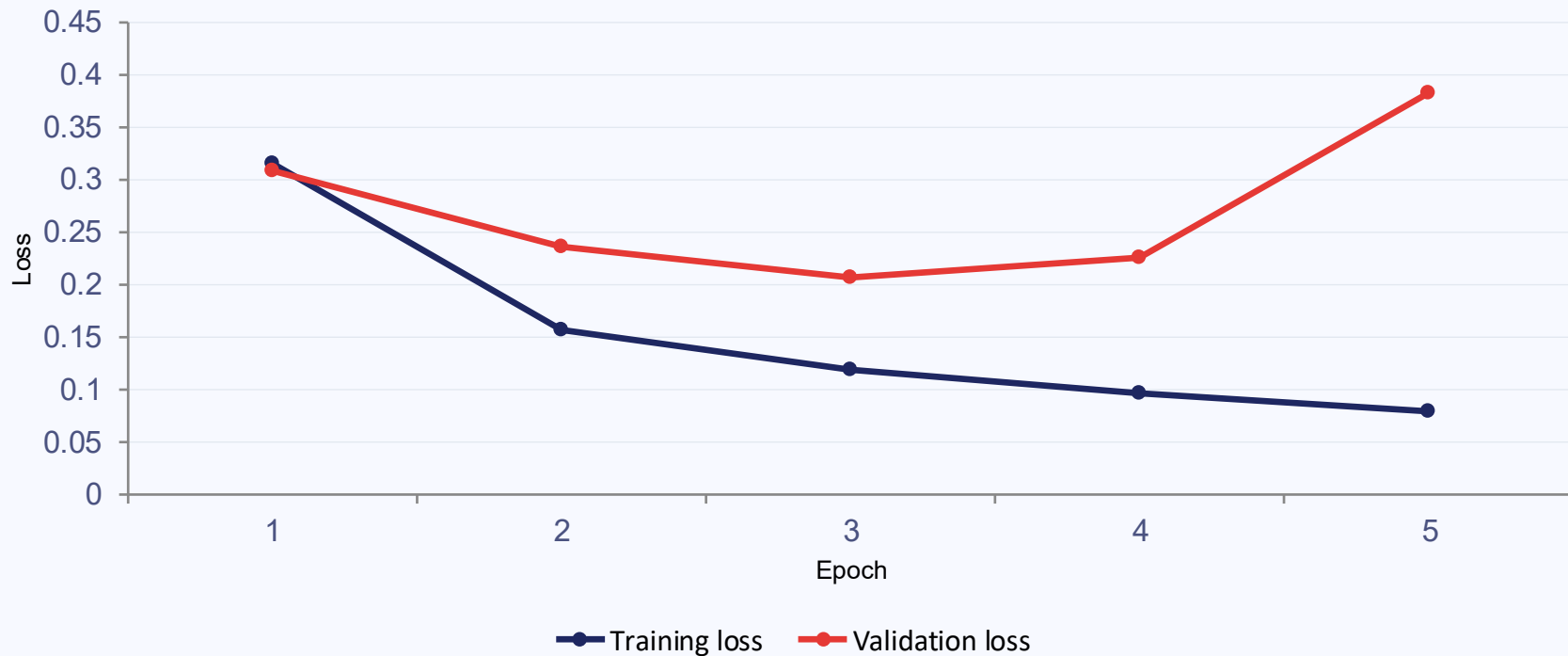
Dataset & Ethics

- **Source**
 - Kaggle — mafiosoquasar/deepfake-faces (DFDC-derived face crops): 86,655 real images and 120,769 fake images.
- **Sampling & split**
 - 5,000 images per class sampled with fixed seed 42; split 70 / 15 / 15 → 7,000 train · 1,500 val · 1,500 test. Split manifest: data/working/manifest.csv.
- **Ethics**
 - Supervisor sign-off: Dr Nicholas Costen (EAN.txt). No children present; no re-identification attempted; no raw face data redistributed in the submission archive.

Implementation — Training

- **Backbone**
 - ResNet18, ImageNet-pretrained (ResNet18_Weights.IMAGENET1K_V1). Final fully-connected layer replaced with `nn.Linear(512, 2)`. 11,177,538 total parameters; all trainable.
- **Optimiser & schedule**
 - AdamW, learning rate $1e-4$, weight decay $1e-4$. Cross-entropy loss applied to raw logits (no softmax in the model). 5 epochs; best checkpoint selected on validation accuracy.
- **Hardware & reproducibility**
 - Trained on Apple M4 (MPS backend) in under 10 minutes. Seed 42 applied across Python, NumPy, and PyTorch. Minor MPS non-determinism noted; CPU runs are fully deterministic.

Training Curves



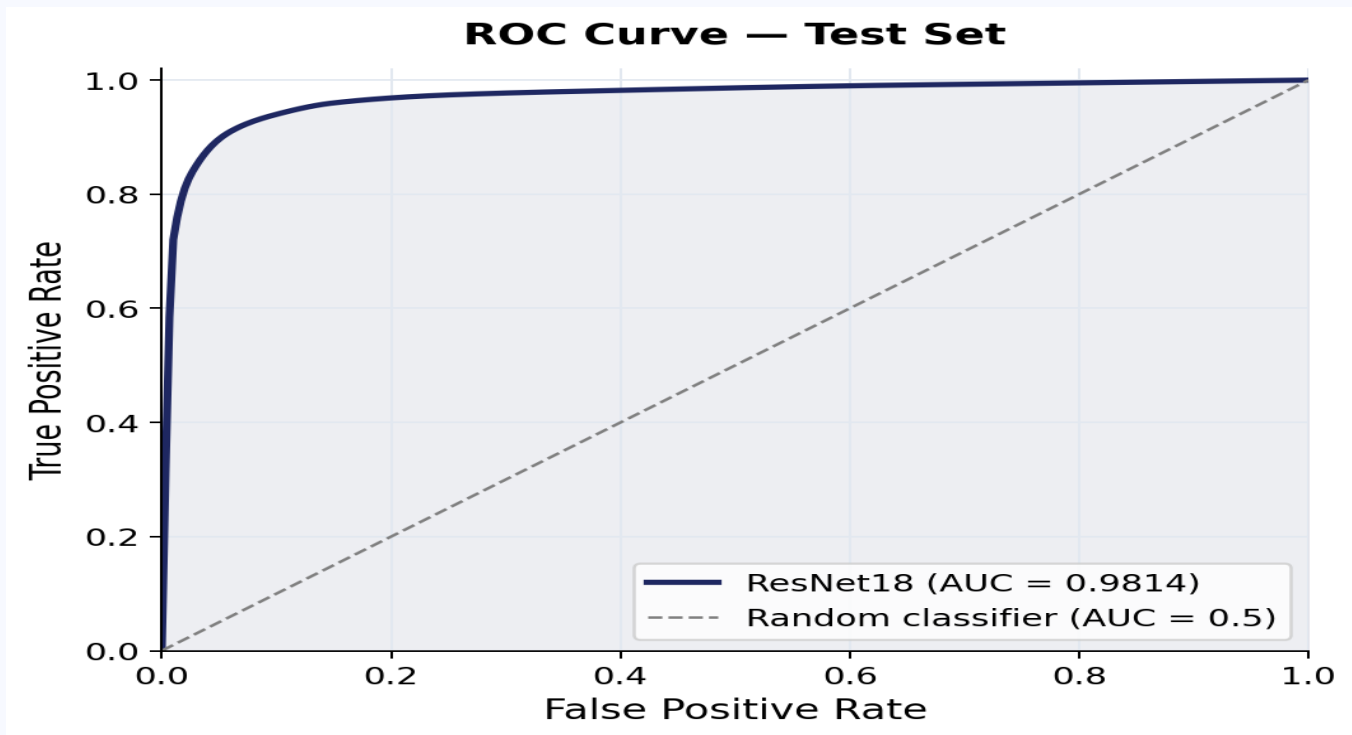
Validation loss diverges from training loss after epoch 3 — best-model checkpointing on val accuracy mitigated overfitting.

Evaluation — Headline Metrics

Metric	Value
Accuracy	0.9247
Weighted F1	0.9246
AUC	0.9814
Test samples	1,500

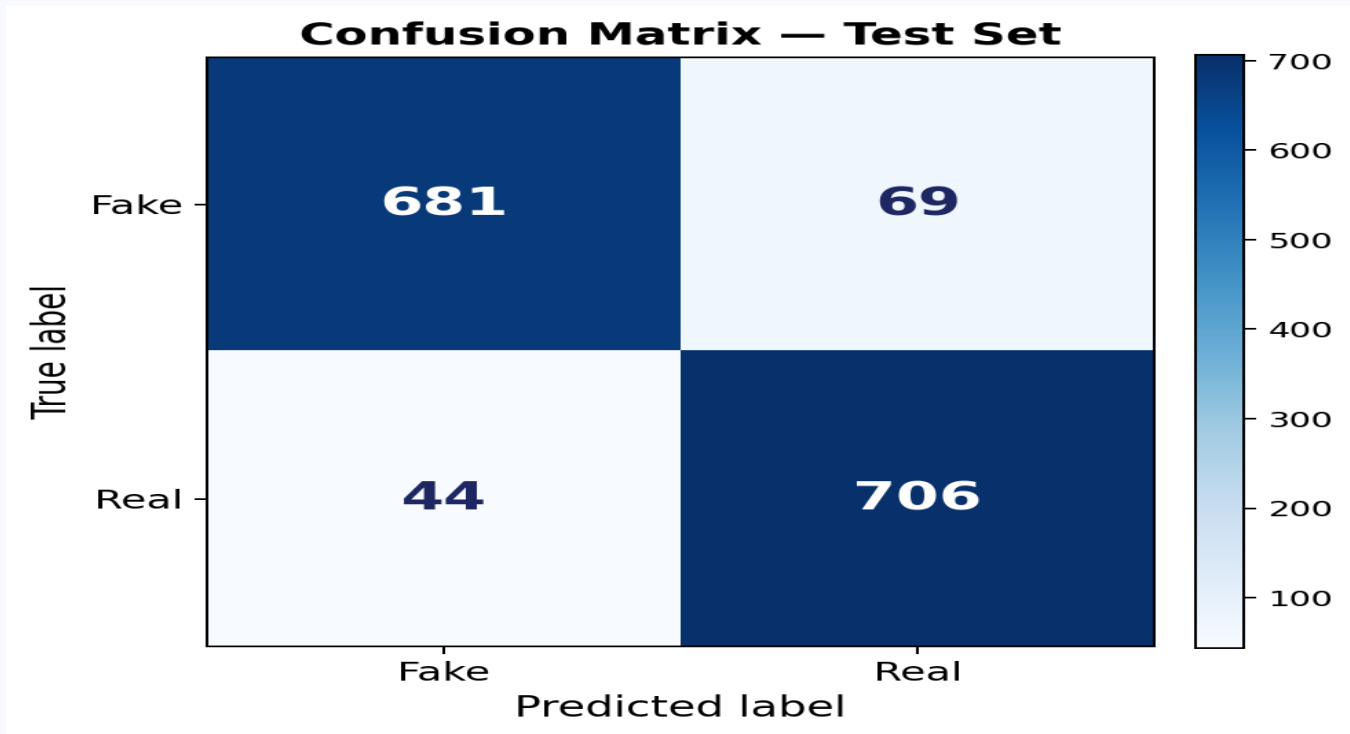
Held-out test set: 750 real + 750 fake face crops.

ROC Curve



AUC = 0.9814 — strong threshold-independent discrimination between real and fake face crops.

Confusion Matrix



Model exhibits a slight false-positive bias — recall on fake class = 0.908.

Limitations & Design Choices

- **Domain shift**
 - The model is trained on DFDC-derived face crops. Out-of-distribution clips (e.g. webcam selfies) produced $P(\text{fake}) = 0.510$ — borderline confidence suggesting reduced generalisation.
- **Single-architecture comparison**
 - Only ResNet18 was evaluated within the project timeline. ResNet50 and Xception comparisons are natural next steps and would strengthen ablation evidence.
- **No temporal modelling**
 - Frame scores are aggregated by arithmetic mean, discarding inter-frame consistency entirely. A 3D CNN or LSTM over frame embeddings would likely improve performance on longer clips.

Conclusions & Future Work

- **Achieved 92.47% accuracy / AUC 0.981**
 - Transfer-learned ResNet18 baseline on a held-out test set of 1,500 balanced face crops — strong performance for a frame-level classifier within the project scope.
- **Natural next steps**
 - Cross-dataset evaluation on FaceForensics++ (to assess cross-generator generalisation), temporal modelling (3D CNN or LSTM), and threshold calibration for in-the-wild deployment.
- **Fully reproducible pipeline**
 - `prepare_data.py` → `train.py` → `evaluate.py` reproduces headline numbers from a cold checkout. Seed 42 applied throughout; `manifest.csv` provides a cast-iron split record.